

**Network Coding for Wireless Applications: A
Review**
Project Activity
Wireless and Mobile Network



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Submitted to:

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Dated:

5th Jun, 2025

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EXECUTIVE SUMMARY

Network Coding (NC) has emerged as a transformative paradigm in wireless communication, enabling more efficient, resilient, and scalable networks by allowing intermediate nodes to encode and combine packets instead of merely forwarding them. This consolidated review integrates key insights from five significant research works, offering a multi-dimensional understanding of NC's application across various wireless architectures and systems.

1. ENHANCING WIRELESS NETWORK PERFORMANCE WITH SDN AND NETWORK CODING:

The integration of Software Defined Networking (SDN) with NC marks a pivotal advancement in centralized and programmable network control. By separating the control and data planes, SDN facilitates dynamic routing and resource allocation, while NC further optimizes performance under high-traffic conditions. Experimental findings demonstrate a 42.8% increase in throughput and reduced packet loss when NC is integrated into SDN frameworks. This synergy promises highly adaptable, fault-tolerant future networks.

2. NETWORK CODING IN PEER-TO-PEER(P2P) CONTENT DISTRIBUTION:

A systematic analysis of 41 studies shows that NC significantly improves throughput and reduces load in P2P content delivery systems. By enabling simultaneous data distribution across multiple paths, NC enhances redundancy and fault resilience. However, computational overhead and block scheduling complexity remain open challenges. The study emphasizes the need for standardized evaluation metrics and coding libraries to facilitate broader adoption and cross-platform optimization.

3. SECURITY IN NC-ENABLED MOBILE SMALL CELLS:THE IDLP FRAMEWORK:

Security remains a critical concern in NC-based systems due to potential pollution attacks—malicious data injection that propagates rapidly. The proposed Intrusion Detection and Location-aware Prevention (IDLP) mechanism addresses this by using a null-space homomorphic MAC scheme to detect and localize malicious nodes. Notably, the system is resource-efficient as it selectively applies security operations, achieving lower communication and computational overhead while enhancing network integrity.

4. PHYSICAL LAYER NC WITH INTELLIGENT REFLECTING SURFACE(IRS):

In the physical layer, integration of intelligent reflective surfaces (IRS) with physical-layer network coding (PNC) opens new frontiers for wireless optimization. In the IRS-Assisted relay system (e.g., butterfly network), the phase-shift optimization bit error rate (BER) significantly increases performance. Analytical and simulation results suggest that such hybrid architecture can dramatically improve the quality and throughput of the signal, represent a novel direction in the next generation (6G) network design.

5. BROADER APPLICATIONS AND CODING SCHEMES IN WIRELESS NETWORKS

The versatility of NC is shown in various types of applications, including file downloads, video streaming, live broadcasting and wireless Aries networks. Various coding strategies - such as random linear network coding (RLNC), triangular network coding (TNC), and opportunistic network coding (ONC).Have been detected to maximize transmission efficiency and reduce packet retreats. While these technologies show obvious advantages on the network layer,

physical-layer implementation remains an active research field with sufficient capacity for adaptation.

LITERATURE REVIEW

Network coding (NC) has become a crucial approach for improving the efficiency of data transmission, throughput, and resilience in wireless networks. Unlike traditional routing methods, NC enables intermediate nodes to manipulate packets algebraically, resulting in more efficient data transmission. This is particularly advantageous in wireless applications, such as wireless sensor networks (WSNs), mobile ad hoc networks (MANETs), vehicular networks (VANETs), and the internet of things (IOT).

Li et al. (2020) suggested a linear network coding approach that was implemented in WSNs to enhance energy efficiency. By encoding data at intermediate nodes and minimizing retransmissions, the network was able to achieve higher throughput while maintaining lower latency.

Zhao & Chen (2021) delved into the concept of adaptive network coding (ANC) for VANETs. Their simulation-based approach demonstrated that ANC improved packet delivery under changing mobility conditions, guaranteeing quality of service in real-time vehicular applications.

Wang et al. (2022) examined the influence of random linear network coding (RLNC) in delay-tolerant networks (DTNs). Their analytical model demonstrated improved resilience to packet loss in networks that experience intermittent connectivity.

Ahmed et al. (2023) examined the use of NC in internet of things (iot) environments, with a focus on its application in smart agriculture. The research combined NC with MQTT protocols, demonstrating improved reliability and energy efficiency.

Kim & Yu (2024) suggested a hybrid approach that combines both non-cooperative and machine learning techniques for wireless mesh networks. Their approach anticipated network states and dynamically adjusted coding strategies, resulting in enhanced throughput and adaptability.

Despite significant progress, obstacles persist, including overhead caused by complex coding, security issues related to encoded data, and the need to balance energy consumption and computational power. Most studies tend to concentrate on theoretical models or simulations, with limited practical implementation and evaluation of their performance in real-world scenarios.

TABLE LITERATURE SUMMARY

Table 1:TABLE LITERATURE OF NETWORK CODING FOR WIRLESS APPLICATIONS:A REVIEW

YEAR	TITLE	SUMMARY (METHODOLOGY)	FINDINGS	STRENGTHS	GAPS/WEAKNESSES	EVALUATED PARAMETERS
2020	Li et al. – “Energy-Efficient Linear Network Coding in WSNs	Model-based method employing linear NC to minimize retransmissions.	Enhanced resource utilization and productivity.	Optimized Code, Minimal Delay.	Expensive computational expense at nodes in between.	Energy consumption, latency, throughput
2021	Zhao & Chen – “Adaptive Network Coding for VANETs”	Adaptive NC with Mobility-Aware Models in NS-3 Simulator.	Improved data transmission rates in dynamic environments.	Dynamic adaptability	Insufficient empirical evidence.	Packet Delivery Rate, Transit Time, Network Effectiveness.
2022	Wang et al. – “RLNC for Delay-Tolerant Wireless Networks”	Computational modeling and Monte Carlo simulations.	Enhanced dependability in networks with limited connectivity.	Robust under packet loss	Performance not evaluated.	Reliability, delay, data excess
2023	Ahmed et al. - "MQTT-based network coding for IOT"	NC integrated with MQTT in Smart Agriculture IOT	Increased reliability and energy savings	Real world IoT application	Limited scalability analysis	Use of energy, reliability, data throughput
2024	Kim and U-"ML-operated NC in Wireless Aries	Hybrid ML-NC approach using reinforcement learning	Adaptive coding enhances the efficiency of data processing.	Guessable adaptability	High overhead and model training complexity	This is about the factors such as throughput, adaptability, and network load.

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